Integrating $\sin^2(x)$ and $\cos^2(x)$ — 3.2

Can't integrate $\sin^2(x)$ by parts:

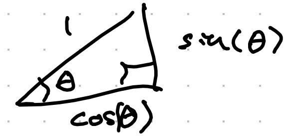
$$\int \sin(x)\sin(x)\,dx \qquad \begin{array}{l} u = \sin(x) \\ dv = \sin(x) \end{array} \Rightarrow \begin{array}{l} du = \cos(x) \\ v = -\cos(x) \end{array}$$

$$= -\sin(x)\cos(x) - \int -\cos^2(x)\,dx \qquad \begin{array}{l} u = \cos(x) \\ dv = \cos(x) \end{array} \Rightarrow \begin{array}{l} du = -\sin(x) \\ v = \sin(x) \end{array}$$

$$= -\sin(x)\cos(x) + \left(\sin(x)\cos(x) - \int -\sin^2(x)\,dx\right) = \int \sin^2(x)\,dx$$

All we're saying is that $\int \sin^2(x)\,dx = \int \sin^2(x)\,dx$

Trig identities



- $\sin^2(x) + \cos^2(x) = 1$ (Pythagorean theorem)

- $\sin(2\theta) = 2\sin\theta\cos\theta$
- $\cos(2\theta) = \cos^2\theta - \sin^2\theta$

$$\cos^2(x) = 1 - \sin^2(x)$$

$$\cos(2\theta) = 1 - \sin^2\theta - \sin^2\theta = 1 - 2\sin^2\theta$$

$$2\sin^2\theta = 1 - \cos(2\theta) \implies \boxed{\begin{aligned} \sin^2\theta &= \frac{1-\cos(2\theta)}{2} \\ \cos^2\theta &= \frac{1+\cos(2\theta)}{2} \end{aligned}}$$

$$\int \sin^2(x)\,dx = \int \frac{1-\cos(2x)}{2}\,dx = \frac{1}{2}\int dx - \frac{1}{2}\int \cos(2x)\,dx$$

$$= \frac{1}{2}x - \frac{1}{4}\sin(2x) + C$$

$$\int \cos^2(x)\,dx = \int \frac{1+\cos(2x)}{2}\,dx = \frac{1}{2}\int dx + \frac{1}{2}\int \cos(2x)\,dx$$

$$= \frac{1}{2}x + \frac{1}{4}\sin(2x) + C$$